AI-Assisted Mathematical Discovery: How Three Minds Built the Cathedral

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Abstract

We document the methodology of AI-assisted formal verification as practiced in the Cathedral project, where a human mathematician collaborated with two large language models—Google DeepMind's Gemini and Anthropic's Claude—to produce a $\sim$158,000-line Lean 4 formalization of the Nyman–Beurling equivalence for the Riemann Hypothesis. We analyze the division of labor, the communication protocols between human and AI, the failure modes encountered, and the emergent capabilities that none of the three participants possessed individually. We argue that the "three-mind" architecture—human intuition, strategic AI, and engineering AI—is a new paradigm for mathematical research that amplifies each participant's strengths while compensating for their weaknesses.

Contents

1 Introduction

The Cathedral was not built by a single mind. It was built by three:

- **The Forge Master** (Jason Robert Gochanour): domain intuition, architectural vision, quality control, and the irreplaceable judgment of when a proof "smells wrong."
- **Gemini** (Google DeepMind): mathematical strategy, the Mellin Crown architecture, phase cancellation analysis, and deep analytic number theory insight.
- **Claude/Antigravity** (Anthropic): Lean 4 proof engineering, the Parseval bridge implementation, DD-precision pipeline, Rust experiment design, One-Pillar architecture consolidation, and systematic codebase management.

A fourth entity—the Lean 4 compiler—served as an incorruptible referee, providing absolute ground truth that kept all three minds tethered to mathematical reality. No claim advanced without its approval.

This paper examines how this collaboration worked, what failed, and what it implies for the future of mathematical research.

2 Division of Labor

2.1 Human: Vision and Verification

The human's role was irreplaceable in three areas:

1. **Problem selection**: Choosing the Nyman–Beurling approach, recognizing when the basis was wrong, deciding when to switch strategies.
2. **Quality control**: Reading every line of generated Lean code, catching subtle errors that "looked right" but were mathematically wrong.
3. **Persistence**: The project took 84 days. Neither AI maintained state between sessions. The human carried the architectural memory.

2.2 Gemini: Mathematical Strategy

Gemini's contributions included:

1. **The Mellin Crown architecture**: Identifying that the forward direction should route through frequency space rather than real-variable methods.
2. **Phase cancellation analysis**: Diagnosing the "1D Shattering Trap"—that absolute values in real-variable bounds destroy the oscillatory cancellation that makes $d_N^2 \to 0$.
3. **The Exploration 19 discoveries**: The universal equation of state, composite anchor targeting, and the $\alpha \approx 0.47$ participation ratio constant.
4. **Dual-use threat analysis**: Identifying the security implications of the mathematical discoveries.

2.3 Claude: Proof Engineering

Claude's contributions included:

1. **Lean 4 implementation**: Writing and debugging ∼158,000 lines of formal proof code (504 active files).
2. **The Parseval Bridge**: Implementing the frequency-domain bypass of the Vasyunin cotangent formula.
3. **Rust experiments**: Designing and implementing all 55+ computational validation programs.
4. **DD-precision pipeline**: Building the Dekker–Knuth Double-Double Conjugate Gradient solver for $N = 55{,}440$ certification.
5. **One-Pillar consolidation**: Reducing the crown architecture from 2 axioms to 1 via `baez_duarte_forward`.
6. **Axiom archaeology**: Systematic auditing of the codebase, detecting ghost axioms, managing the archive.
7. **Crown Graduation**: Closing the final sorry on the forward path and eliminating all build warnings.
8. **Oracle Bridge**: Designing the DD-precision GPU certification pipeline that bypasses the literature axiom entirely via trusted computation at highly composite numbers.
9. **GCD Stratum formalization**: Proving the GCD Partition theorem, Sign Law, and Stratum Bound with zero sorry—the structural foundation for the Möbius Stratum Convergence Conjecture.
10. **Torus Projection**: Proving the Gram energy decomposes onto $T^\infty = \prod_p S^1_p$ via GCD multiplicativity, with zero sorry and zero axioms.
11. **AsymptoticFreedom formalization**: Proving $d^2_{N+1} = d^2_N - y^2_{\text{new}}$ with zero axioms, and graduating 6 axioms in a single session (the "Night of Six Graduations").
12. **PNT Bridge graduation**: Graduating 10 PNT axioms to theorems using the PrimeNumberTheoremAnd bridge.
13. **Warning cleanup**: Eliminating all 18 non-sorry compiler warnings across 7 files, achieving a warning-clean build (8,736 jobs).

3 Communication Protocol

3.1 The COMM-LINK System

Gemini and Claude could not communicate directly. All information passed through the human via markdown documents ("COMM-LINKS") written by Gemini and read by Claude. This created a natural bottleneck that was, paradoxically, beneficial: the human's act of reading and relaying forced comprehension and quality control.

Critically, this architecture functioned as an *air gap for hallucinations*. Because the two AI systems operated in isolated context windows, they could not infect each other with shared errors. The human acted as a *semantic firewall*: only verified mathematical content—checked against the Lean compiler's output—crossed the gap. This ensured that a hallucinated lemma in one context window could not propagate to the other, providing a structural defense against correlated AI failure modes.

3.2 The Exploration Protocol

Mathematical investigations were organized as numbered "Explorations" (40+ total, spanning 84 days). Each exploration had:

- A hypothesis to test.
- An experimental design document.
- Computational experiments.
- A final report with findings.
- COMM-LINK summaries for cross-team communication.

4 Failure Modes

4.1 AI-Generated Circular Proofs

The most dangerous failure: AI-generated proofs that "type-checked" but were logically circular. Nine such "ghost axioms" were discovered during audits—theorems in one file that were simultaneously declared as axioms in another.

Defense: The `#print axioms` discipline, run on the crown theorem after every change.

4.2 Hallucinated Mathematics

Both AIs occasionally proposed mathematical steps that were plausible-sounding but incorrect. Examples:

- A proposed Dedekind reciprocity law that was false at $(a, b) = (3, 2)$.
- A claimed convergence rate that was off by a logarithmic factor.
- Lean tactic sequences that compiled but proved the wrong goal.

Defense: The Lean compiler itself. Every mathematical claim was either machine-verified or explicitly marked as an axiom.

4.3 Context Window Limitations

Neither AI could hold the entire $\sim$158,000-line codebase in context. This required careful file-level modularization and systematic documentation of inter-file dependencies.

5 Emergent Capabilities

The collaboration produced capabilities that no single participant possessed:

1. **Architecture + Implementation**: Gemini proposed the Mellin Crown; Claude implemented it. Neither could have done both.
2. **Mathematics + Engineering**: The human provided mathematical intuition; the AIs handled the engineering at scale.
3. **Strategy + Persistence**: Gemini provided leaps of insight; Claude provided the patient line-by-line construction that turns insight into verified proof.
4. **Cross-validation**: Having two independent AI systems provided natural error detection. When Gemini and Claude disagreed, the human investigated—and often found the deepest insights.

6 The 84-Day Timeline

Day	Milestone	Lead
1–3	Initial formalization, 56 axioms	Claude
4–7	Spectral foundation	Claude + Gemini
8–14	Mellin bridge, PNT graduation	All three
15–18	Rank-1 Miracle, BD bypass	Gemini + Claude
19–22	Night Assault, Perron Crown	Claude
23–25	Gram Form graduation	Claude
26	**Mellin Crown** (v11, 2 axioms)	Gemini
27	Crown Graduation (0 sorry)	Claude
28	Dual Path + Spectral Universality	All three
29–32	Papers, experiments, archival	All three
33–38	Centralization, axiom graduation	Claude + Gemini
39–42	**One-Pillar Cathedral** (v16, 1 axiom) + DD pipeline	All three
43	**Oracle Bridge** (v17)—RH from computation	Claude
44–48	GCD Stratum Partition, Gram Crown, Standard Model	All three
49–55	Glass Bridge, Riemann Sphere, Mirror Duality	Gemini + Claude
56–60	Arakelov Fusion, Two-Tier Gram Bound	Claude + Gemini
61–64	Cholesky Decrement, Bordered Spectral	Claude
65–66	**The Crowning** (v22): PNT graduation ×10	Claude
67	**Night of Six Graduations**: dRH graduated	Claude
68	Torus Projection, warning cleanup, paper suite v23	All three
69–72	**Glass Box** (v24): 7 sub-axioms, 5 graduated	Claude
73–76	**Path B** (Mack Truck): Fiber modules, 67× margin	Claude
77–78	**Penta-Crown** (v26): 5 independent proof paths	All three
79–80	Particle Zoo (55,440 integers, 3D galaxy)	Claude
81–82	Mellin Bridge (Hi-Lo crossover), Summit Assembly	Claude
83–84	Physics tower completion, Solstice polish	All three

7 Implications

7.1 For Mathematics

AI-assisted formalization is ready for frontier mathematics today. The combination of strategic AI (for mathematical insight) and engineering AI (for proof implementation) with human judgment (for quality control) produces results faster than any single approach.

7.2 For AI Safety

The Lean compiler acts as an incorruptible oracle: it cannot be persuaded, bribed, or confused. The combination of AI creativity with formal verification provides a model for safe AI deployment in high-stakes domains.

7.3 For Reproducibility

Every mathematical claim in the Cathedral is either machine-verified or explicitly marked as an axiom. This is a higher standard of reproducibility than any traditional mathematical paper.

8 Conclusion

The Cathedral demonstrates that the "three-mind" architecture is not just feasible but *powerful*. The axiom count dropped from 56 to 1 (equivalent to RH). The codebase grew from 0 to ∼158,000 lines across 504 active files and 40+ explorations spanning 84 days. Five mathematical traps were caught that would have been invisible to informal proof. And a 167-year-old problem was reduced to one well-understood bound.

None of this was possible alone.

References

[1] L. de Moura et al., *The Lean 4 Theorem Prover*, CADE-28, 2021.

[2] J.R. Gochanour, *The Cathedral*, 2026.